FLEET TEAM DIGEST

737MAX-FTD-35-24002

Issue Title : AV20145 Series Portable Oxygen Cylinders Leaking in Service When Installed on Airplanes Parked at Low Temperature Below 0 deg C

Airplane Model	ATA	Minor Model(s)
737MAX	3531-00	-8, -9
Other Model(s)	**ECCN**	
MD-11, 777, 747-400, 737	9E991	

Originated Date	Last Revised Date	Created On
12/20/2024	08/07/2026	07/22/2026
Estimated Completion Date	**Next Update Date**	
07/25/2029	12/16/2027	

Revision Description

Milestone dates revised.

Collins' new portable cylinder design and qualification scope includes requirements from Boeing and several other original equipment manufacturers (OEMs). Collins requires additional time to internally coordinate and align these requirements.

Applicability

Airplanes with Collins Aerospace Integra AV20145 Series Portable Oxygen Cylinders (POCs) installed.

Description

The POC supplier Collins Aerospace (formerly BE Aerospace) has informed Boeing of several operators reporting leaking Integra AV20145 Series POCs installed on airplanes that have been parked in low temperature below 0 deg C conditions. These POCs were reported to have lost pressure in a relatively short amount of time.

Background

The AV20145 Series POCs were originally only installed on other non-Boeing Original Equipment Manufacturer (OEM) airplanes. Due to European customers' desire to use the Collins Aerospace POCs, Boeing qualified and installed these POCs on 777 airplanes starting in 2016. Boeing has also installed these POCs on 737-8 and 737-9 airplanes per customer desire. Boeing is also aware of these POCs being installed by customers post-delivery via third party Supplemental Type Certificates (STCs) on 737-300, 737-500, 747-400 and MD-11F airplanes.

Various customers have informed Collins Aerospace that on aircraft that have been parked in cold destinations, the Integra AV20145 series portable oxygen cylinders showed occasional pressure losses at low temperature below 0 °C in a relatively short time.

In response to the customer reports, Collins Aerospace requested the affected portable cylinders to be returned from the respective customers to investigate under what low temperature conditions leakage occurred and determine the root cause of the leaks at low temperatures.

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After the equipment from the customers was received, and then charged to the operating pressure of 1850 psi and stored at different temperatures beginning from +20 °C to -40 °C. The investigation showed that minor leakage could begin after the cylinders being exposed to a temperature of -10 deg C for 15 hours. Cylinder pressure could drop to 0 psi after exposing to a temperature of -20 deg C for 4 hours.
The leakage occurred at the Integra Valve, specially at the interface between pressure gauge and valve lower body.
The cause of the leakage is due to the internal valve sealing design and the material compatibility at low temperature which limit the cylinder performance at low temperature, leading to leakage and pressure loss.

Status

Collins Aerospace is working to develop the new POCs to replace the AV20145 Series POCs for addressing the issues of the leakage at the low temperatures, enhance its reliability and performance. The improved POCs will have the new part numbers (P/Ns) and the new P/Ns will require certification.

Interim Action

None

Final Action

Boeing will work with Collins Aerospace to certify the new POC P/Ns and establish the applicable interchangeability.

Boeing expects Collins Aerospace to publish a service bulletin to modify the existing POCs to the new configuration. Boeing will publish a service letter to inform operators of the new POC P/Ns and interchangeability once available.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 06-NOV-2023 (Complete)

Solution Selected: 01-DEC-2023 (Complete)

Change Committed: 4th Q 2027 (Target)

Parts Available: 1st Q 2029 (Target)

Production Incorporation: 2nd Q 2029 (Target)

Collins Service Bulletin Issued: 2nd Q 2029 (Target)

Boeing Service Letter Issued: 2nd Q 2029 (Target)

Operator Action

Boeing recommends that operators follow the Flight Attendant Manual (FAM) portable oxygen bottle preflight check prior to

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airplane dispatch.

If the airplanes are to be parked in extremely low temperatures, operators should be aware that these POCs may lose pressure.

Prior to parking in such conditions, operators may choose to remove POCs and store them in a warmer location and then re-install prior to resuming operation.

Boeing notes that there is no concern with the POCs losing pressure during flight if they have passed the preflight check because the passenger cabin temperature is well above the noted extremely low temperature where the leaks were reported.

Supplier Information

Collins Aerospace (B/E Aerospace) CAGE Code D1379

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Portable Oxygen Cylinder Assembly		Collins Aerospace	AV2014595-11
Portable Oxygen Cylinder Assembly		Collins Aerospace	AV2014595
Portable Oxygen Cylinder Assembly		Collins Aerospace	AV2014595-11-2
Portable Oxygen Cylinder Assembly		Collins Aerospace	AV2014586-11-2

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations

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